Multimodal Autonomous Rover for Precision Agricultural Analysis

EE175B-022
Group 24
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Design Concept and Application

Project Design

Remote-operated/autonomous rover that performs mapping, navigation, and plant analysis through the implementation of LiDAR, motor control, computer vision, and long-range wireless communication.

Purpose and Meaning

- Automation of Crop Maintenance
  - Reduce need for manual crop inspection in large agricultural fields
- Precision Agriculture
  - Collect accurate plant data which can be used to optimize resource allocation
- Research of Intelligent Systems
  - Integrate multiple technologies into a complete and operational system

Technical Principles

- Embedded Systems
  - Microcontroller architecture
  - GPIO digital interfacing
  - UART serial communication
- Digital Communication
  - Signal encoding
- Sensor Integration
  - Data acquisition
  - Analog-to-Digital Conversion
- Power Electronics
  - Voltage regulation
  - Efficiency optimization

Intended Applications

- Remote agriculture monitoring
  - Long-range wireless communication
  - Isolated network environments where stable Wifi is not available.
- Precision plant analysis
  - Inspect crops, detect disease, track growth
- Autonomous field mapping
  - Map generation of complex environments

Technical Design Objectives

System Requirements

- **Hybrid Navigation**
  - Remote-operated manual driving
  - LiDAR-based SLAM for autonomous global navigation
- **Image Capture and Plant Analysis**
  - Rover detects orange flag to capture image
  - JPEG file is detected and analyzed through PlantCV
- **Telemetry Communication**
  - Broadcast telemetry data to the base station using LoRa wireless communication
    - Battery voltage
    - Navigation status
    - Plant health percentage
    - Rover position
- **Rover Reliability**
  - Continuous operation on a single battery charge
  - Halt of movement in the occurrence of LoRa disconnection or “STOP” command received
  - Log operational data for post-expedition analysis

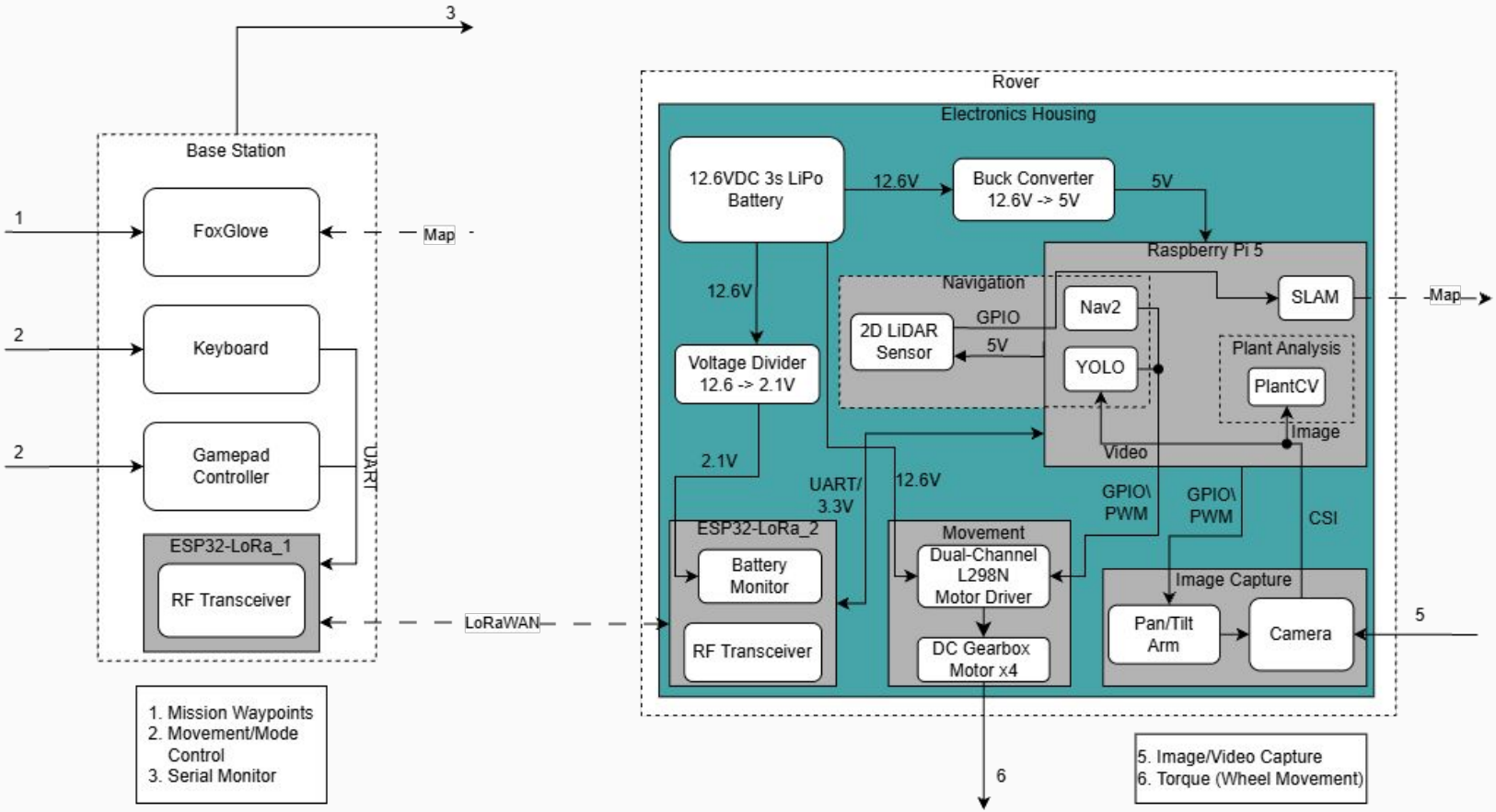
Hardware

- Power
  - 3S LiPo Battery: 9-12.6V operating range
  - Buck Converter: ~5V for Raspberry Pi 5
- Sensors
  - LiDAR scanning frequency: ≥ 5 Hz
- Motors
  - Minimum rover speed: ≥ 0.5 m/s

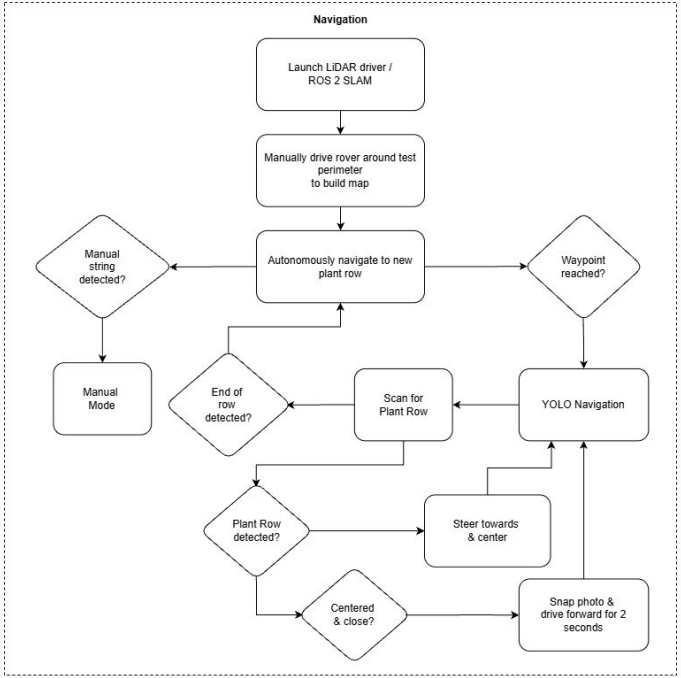
Software

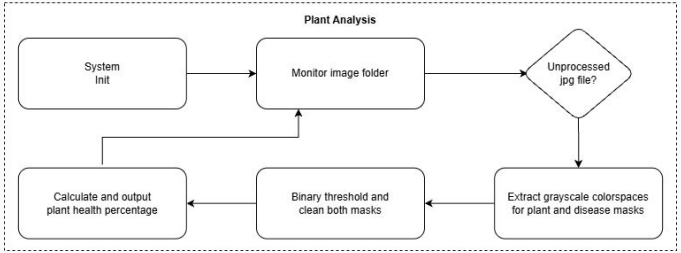
- Nav2
  - Path planning computation time of ≤ 500ms
- YOLO
  - Flag detection accuracy of ≥ 80%
- PlantCV
  - Overall health percentage within 30% of expected value
- LoRa
  - Telemetry packet success rate of ≥ 95%

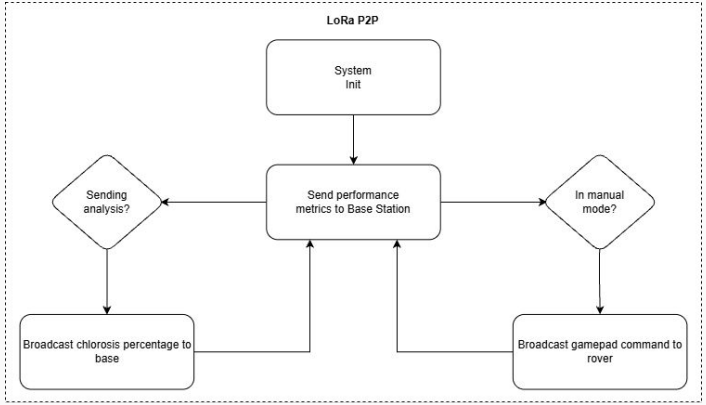
System Block Diagram

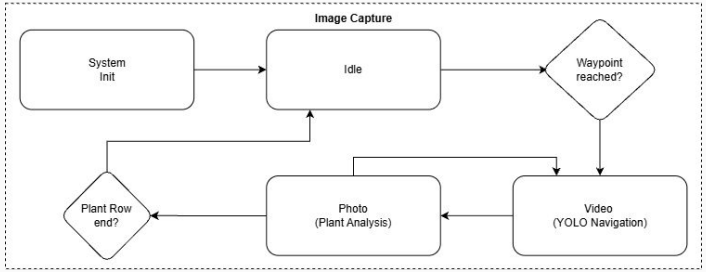


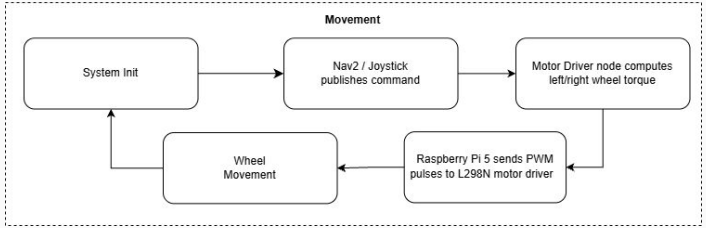
Subsystem Flow Charts





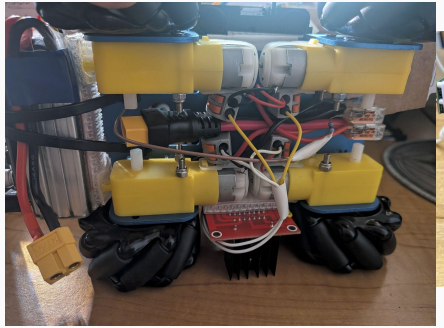


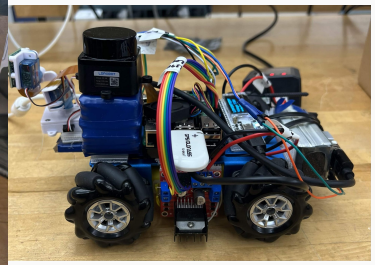




Technical Challenges

Hardware

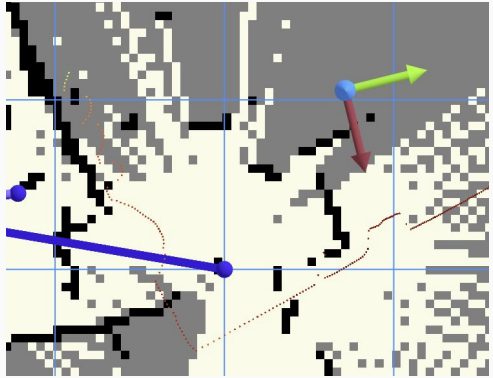




- Rover's low height required a up/down and left/right sweeping motion to track plants.
- Pan/Tilt unable to synergize with camera detection.

- TT motors vulnerable to stripped internal gears under load and heavy acceleration.
- Battery monitor gives false reading if voltage divider is dislodged from power junction.

Software



- **PlantCV**
  - Contrast between plant leaves and discoloration
  - Variable lighting in different environments
  - Computation can jam other CPU processes.
- **YOLO**
  - Camera unable to efficiently track plant marker. Requires RGB threshold tuning.

- **FoxGlove**
  - Slam map visualization required careful TF tree configuration.
  - Waypoints depend heavily on map stability.
- **ROS2 Humble**
  - Autonomous navigation fails if the SLAM map is not properly built.
  - State switching fails if the proper topics are not being published.

Major Design Components and Implementation

Chassis Assembly	Angel	Complete	• Assembled metal frame, 3-6V TT DC motors, and omni-directional wheels.
YOLO / Camera	Ying		RF2O & Navigation Pan-Tilt Gimbal Vision & Auto-Capture Autonomous exploration
PlantCV	Adam		• Percentage output through analysis of plant and disease masks
LoRA / Motor Driver	Angel		• LoRA modules configured for P2P communication. • L298N motor driver
LiDAR	Ying/Angel		• Obstacle detection for autonomous navigation

Design Considerations

Project Constraints

- **Cost Constraints**
  - ~$150 budget for all components
  - Low-cost sensors reduce precision and range compared to advanced hardware
- **Power Consumption**
  - Voltage must be regulated from the 12.6V battery to operate the Raspberry Pi
- **Weight and Size**
  - 6.69″ x 5.51″ Chassis limits placement of devices on the rover
- **Time Constraints**
  - Existing robotics frameworks were used to reduce development time such as ROS2, Nav2, and SLAM

Industry Standards

- **LoRa Wireless Communication**
  - Transmits telemetry data and manual commands rather than high-bandwidth video streams
- **UART Serial Communication**
  - Simplifies integration between embedded devices
  - Limits maximum data throughput
- **GPIO and PWM Control**
  - Operation of motors using signals from the Raspberry Pi
  - Precise speed and position control using limited number of available pins

Test Report

LoRa Communication

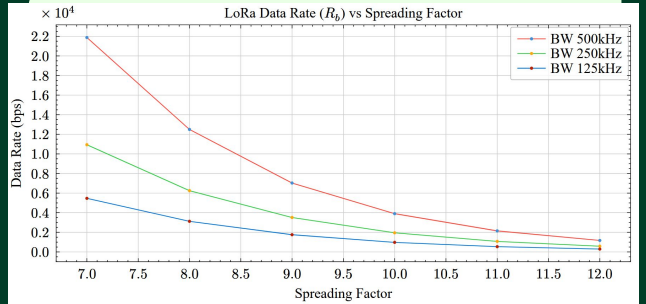


Figure 1: Plot of values of R_b for various values of BW and SF.

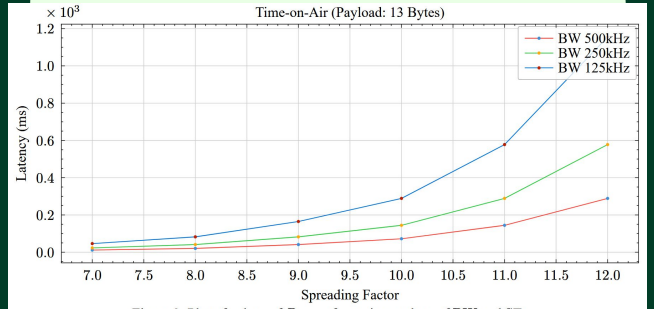
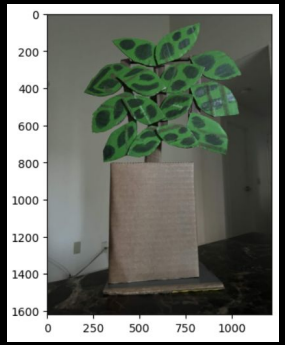
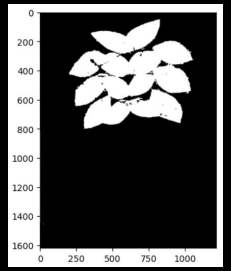


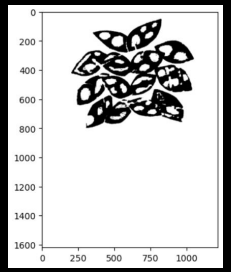
Figure 2: Plot of values of $P_{payload}$ for various values of BW and SF.

- Maximize data rate R_b.
- Minimize time-on-air T_packet.

PlantCV Analysis



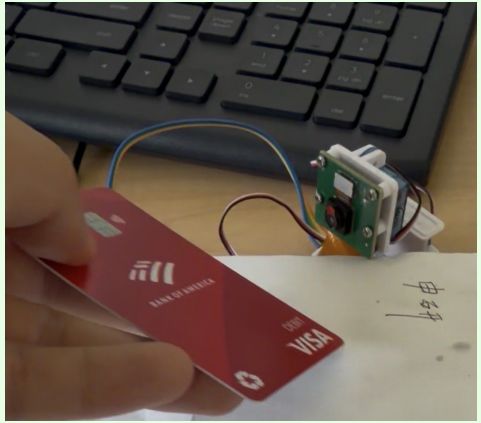


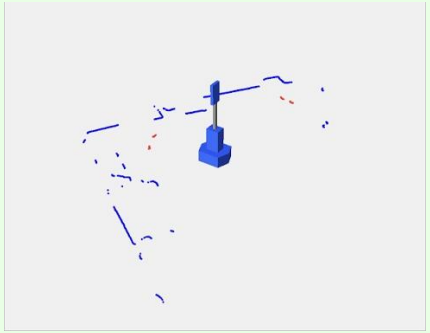


Overall Plant Health: 50.51%

- Simulated using a model plant
- Performed within ~15% of expected value
- Achieved design specifications

YOLO / LiDAR



- 

Demonstration

Summary

Our senior design project aims to develop an autonomous 4WD rover to monitor crop health. The rover is equipped with a pan/tilt camera and an onboard computer that utilizes an open-source machine learning (ML) algorithm to analyze plant images. The processed health data is then transmitted wirelessly to a base station monitor.

To enable autonomous navigation, a user must first manually pilot the rover via a gamepad controller. During this initial manual phase, onboard 2D LiDAR and GPS data are used to construct a SLAM map of the environment and establish navigational waypoints. Once mapped, the rover uses this environmental "memory" to navigate the area autonomously. When approaching target crops, a secondary ML object detection algorithm guides the rover along plant rows by identifying corners and flower beds. If needed, the base station can override the autonomous system at any time to return manual control to the gamepad.

Thank you.

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